reproducible_simulation

Download the necessary packages.

```
library(Hmisc) # for data wrangling
library(mice) # for missing data imputation
library(tidyverse) # for data wrangling
```

Import the data files and merge them to one data frame.

```
# import the raw data
d1 <- sasxport.get("data/DEMO_I.xpt")
d2 <- sasxport.get("data/BPX_I.xpt")
d3 <- sasxport.get("data/BMX_I.xpt")
d4 <- sasxport.get("data/GHB_I.xpt")
d5 <- sasxport.get("data/TCHOL_I.xpt")

# select relevant variables
d1.t <- subset(d1,select=c("seqn","riagendr","ridageyr"))
d2.t <- subset(d2,select=c("seqn","bpxsy1"))
d3.t <- subset(d3,select=c("seqn","bmxbmi"))
d4.t <- subset(d4,select=c("seqn","lbxgh"))
d5.t <- subset(d5,select=c("seqn","lbdtcsi"))

# merge the variables into 1 data frame
d <- merge(d1.t,d2.t)
d <- merge(d,d3.t)
d <- merge(d,d4.t)
d <- merge(d,d5.t)
```

Rename the variables.

```r
# rename variables:
# RIAGENDR - Gender
# RIDAGEYR - Age in years at screening
# BPXSY1 - Systolic: Blood pres (1st rdg) mm Hg
# BMXBMI - Body Mass Index (kg/m**2)
# LBDTCSI - Total Cholesterol (mmol/L)
# LBXGH - Glycohemoglobin (%)
d$age <- d$ridageyr
d$sex <- d$riagendr
d$bp <- d$bpxsy1
d$bmi <- d$bmxbmi
d$HbA1C <- d$lbxgh
d$chol <- d$lbdtcsi
d$age[d$age<18] <- NA
```

Remove NA values.

```r
# select complete cases:
dc <- cc(subset(d,select=c("age","sex","bmi","HbA1C","bp")))
```

Linear regression analysis.

```r
# analysis:
summary(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

```
Call:
lm(formula = bp ~ HbA1C + age + as.factor(sex), data = dc)

Residuals:
Systolic: Blood pres (1st rdg) mm Hg
    Min      1Q  Median      3Q     Max
-49.887 -10.509  -1.378   8.491 107.583

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     98.75149    1.21418  81.332  < 2e-16 ***
HbA1C            1.12638    0.20291   5.551 2.98e-08 ***
age              0.44486    0.01284  34.648  < 2e-16 ***
as.factor(sex)2 -3.24792    0.45164  -7.191 7.34e-13 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.1 on 5088 degrees of freedom
Multiple R-squared:  0.2305,    Adjusted R-squared:   0.23
F-statistic:   508 on 3 and 5088 DF,  p-value: < 2.2e-16
```

```
confint(lm(bp ~ HbA1C + age + as.factor(sex), data=dc))
```

```
                     2.5 %      97.5 %
(Intercept)     96.3711755 101.1317982
HbA1C            0.7285836   1.5241825
age              0.4196932   0.4700355
as.factor(sex)2 -4.1333281  -2.3625106
```

```
summary(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

```
Call:
lm(formula = bp ~ HbA1C + bmi + age + as.factor(sex), data = dc)

Residuals:
Systolic: Blood pres (1st rdg) mm Hg
    Min      1Q  Median      3Q     Max
-51.068 -10.251  -1.504   8.264 107.410

Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     92.65583    1.39320  66.506  < 2e-16 ***
HbA1C            0.75177    0.20596   3.650 0.000265 ***
bmi              0.28632    0.03282   8.724  < 2e-16 ***
age              0.44586    0.01275  34.979  < 2e-16 ***
as.factor(sex)2 -3.63115    0.45049  -8.060  9.4e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 15.98 on 5087 degrees of freedom
Multiple R-squared:  0.2418,    Adjusted R-squared:  0.2412
F-statistic: 405.7 on 4 and 5087 DF,  p-value: < 2.2e-16
```

```r
confint(lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc))
```

```
                     2.5 %     97.5 %
(Intercept)     89.9245592 95.3871089
HbA1C            0.3479966  1.1555348
bmi              0.2219815  0.3506673
age              0.4208695  0.4708464
as.factor(sex)2 -4.5143014 -2.7479929
```

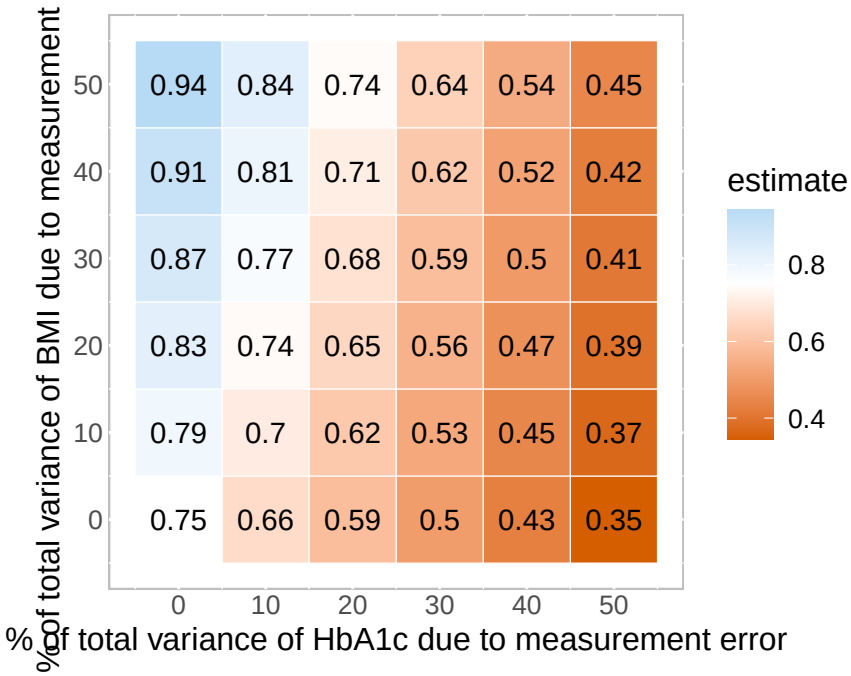
Set up a simulation for the influence of measurement error on the estimate of the beta coefficient for HbA1C in a linear regression model.

```r
# set seed
set.seed(123)

# real coefficent for HbA1C
ref <- lm(bp ~ HbA1C + bmi + age + as.factor(sex), data=dc)$coef[2]

# number of simulations
n.sim <- 1000

# set up different variance scenarios from .0 to .5 per .1 increment
scenarios <- expand.grid(seq(0,.5,.1),seq(0,.5,.1))

# actual variance of HbA1C
var.exp <- var(dc$HbA1C)

# actual variance of bmi
var.conf <- var(dc$bmi)
n <- dim(dc)[1]
beta.hat <- matrix(ncol=dim(scenarios)[1], nrow=n.sim)
# loop over n simulations
for (k in 1:n.sim){
  for (i in 1:dim(scenarios)[1]){
    # compute new variance
    var.me.exp <- var.exp*scenarios[i,1]/(1-scenarios[i,1])
    var.me.conf <- var.conf*scenarios[i,2]/(1-scenarios[i,2])
    # resample data based on new variance (and normality)
    dc$HbA1C.me <- dc$HbA1C + rnorm(dim(dc)[1], 0, sqrt(var.me.exp) )
    dc$bmi.me <- dc$bmi + rnorm(dim(dc)[1], 0, sqrt(var.me.conf) )
    # perform regression analysis based on new variance
```

```
    beta.hat[k,i] <- lm(bp ~ HbA1C.me + age + bmi.me + as.factor(sex), data=dc)$coef[2]
  }}
```

Create a figure for the different combinations of variance for BMI and HbA1c and the resulting beta coefficient obtained from the linear model.

```
# create figure:
tot.mat <- cbind(100*scenarios,apply(beta.hat,2,mean))
colnames(tot.mat) <- c("me.exp","me.conf","estimate")
FIGURE <- ggplot(tot.mat, aes(me.exp, me.conf)) +
  geom_tile(color="white",aes(fill = estimate)) +
  geom_text(aes(label = round(estimate, 2))) +
  scale_fill_gradient2(low="#D55E00",mid="white",high = "#56B4E9", midpoint=ref) +
  labs(x=paste("% of total variance of HbA1c due to measurement error"),
       y=paste("% of total variance of BMI due to measurement error")) +
  coord_equal()+
  scale_y_continuous(breaks=unique(tot.mat[,1]))+
  scale_x_continuous(breaks=unique(tot.mat[,1]))+
  theme(panel.background = element_rect(fill='white', colour='grey'),
        plot.title=element_text(hjust=0),
        axis.ticks=element_blank(),
        axis.title=element_text(size=12),
        axis.text=element_text(size=10),
        legend.title=element_text(size=12),
        legend.text=element_text(size=10))
FIGURE
```



What we see is that, when the variance of HbA1c increases, the beta coefficient tends to be lower. But as the variance of BMI increases, the beta coefficient for HbA1c tends to be higher. There is probably a high correlation between both variables, and as one of the two becomes more variant, that usually highers the beta coefficient for the other variable.